Circuit Theory

Chapter 10

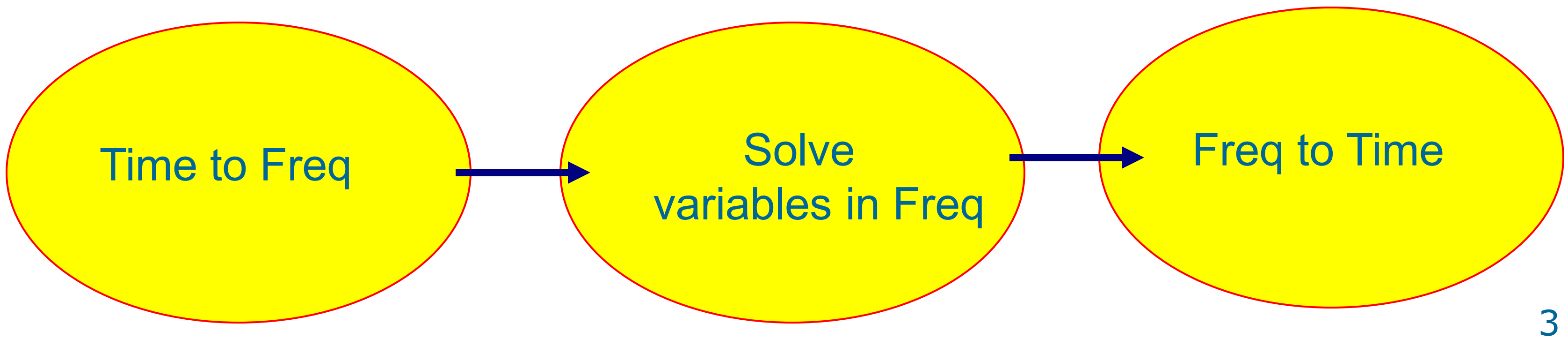
Sinusoidal Steady-State Analysis

Sinusoidal Steady-State Analysis
Chapter 10

10.1 Basic Approach (1)

Steps to Analyze AC Circuits:

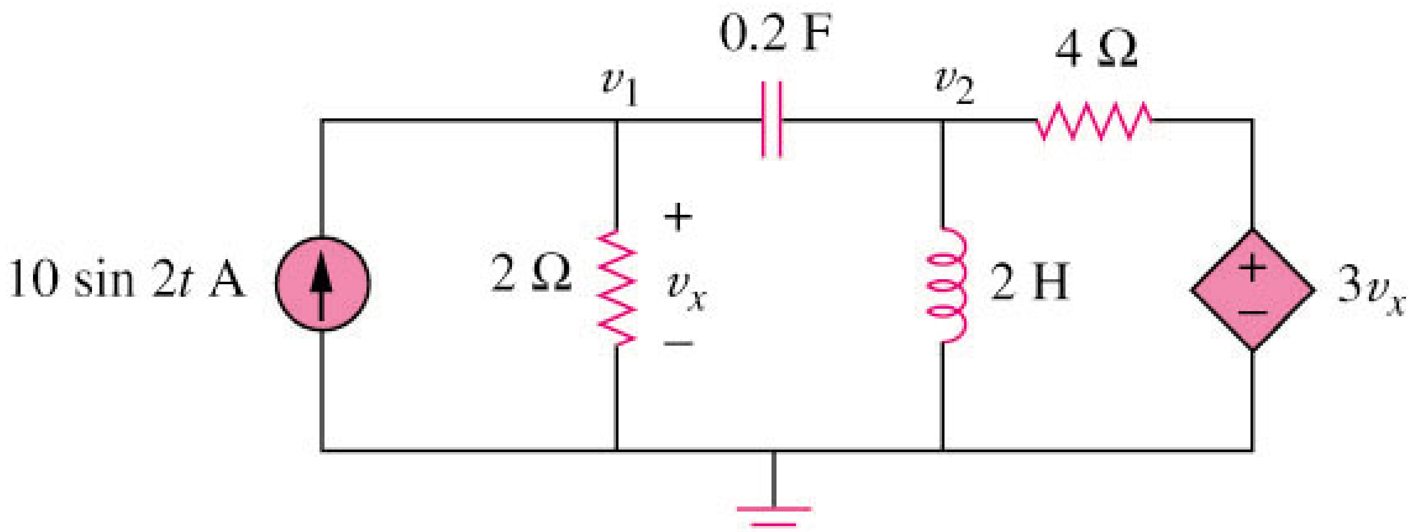
1. Transform the circuit to the phasor or frequency domain.
2. Solve the problem using circuit techniques (nodal analysis, mesh analysis, superposition, etc.).
3. Transform the resulting phasor to the time domain.



10.2 Nodal Analysis (1)

Example 1

Using nodal analysis, find v_1 and v_2 in the circuit of figure below.



Answer:

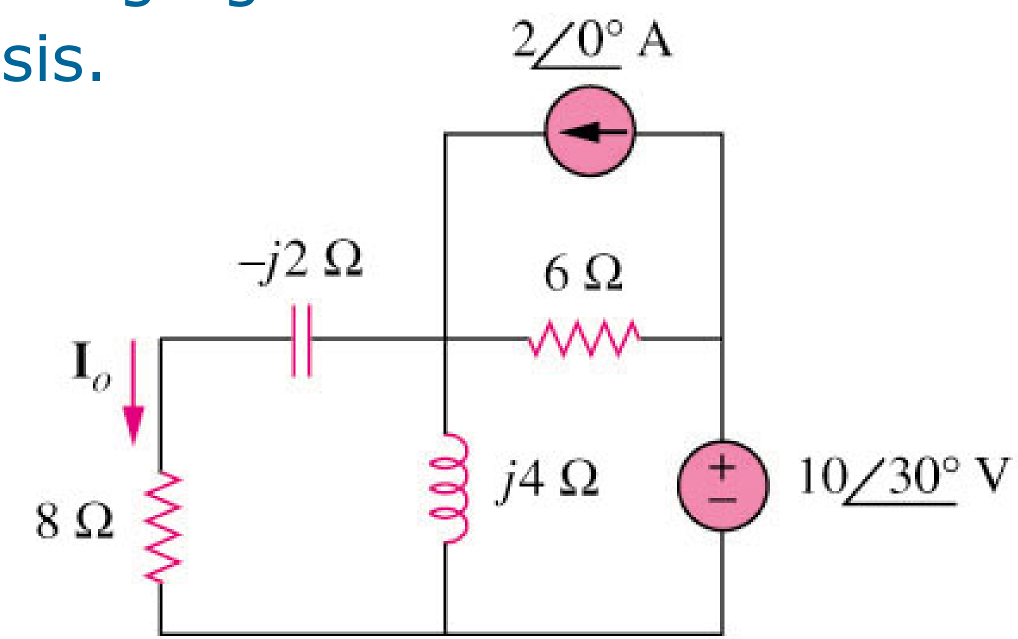
$v_1(t)$ = **11.32 sin(2t + 60.01°) V**

$v_2(t)$ = 33.02 sin(2t + 57.12°) V

10.3 Mesh Analysis (1)

Example 2

Find I_o in the following figure using mesh analysis.



Answer: $\mathbf{I_o} = 1.194\angle 65.44^\circ$ A

HW10 Ch10: 1, 7, 25, 28

10.4 Superposition Theorem (1)

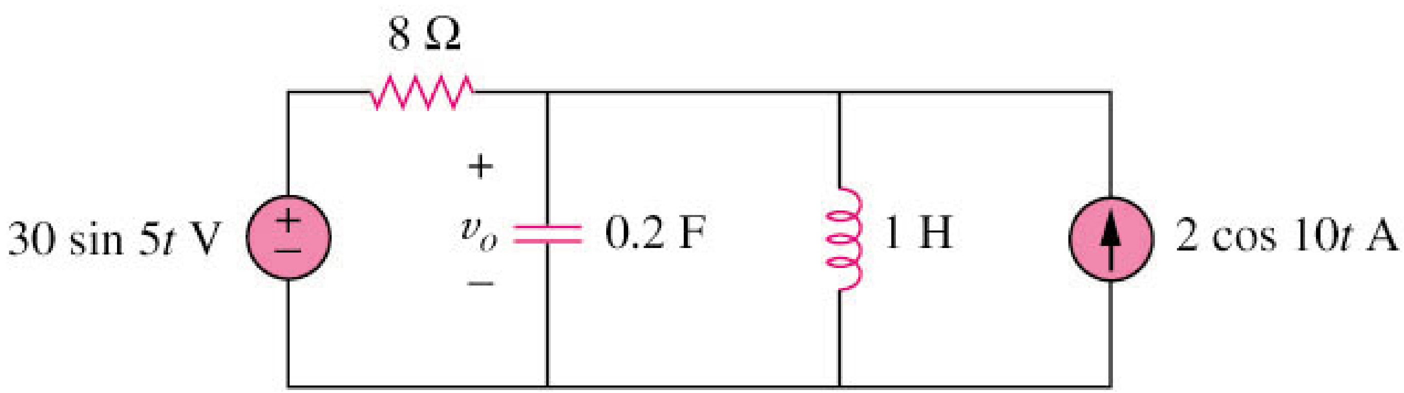
When a circuit has sources operating at different frequencies,

- The separate phasor circuit for each frequency must be solved independently, and
- The total response is the sum of time-domain responses of all the individual phasor circuits.

10.4 Superposition Theorem (2)

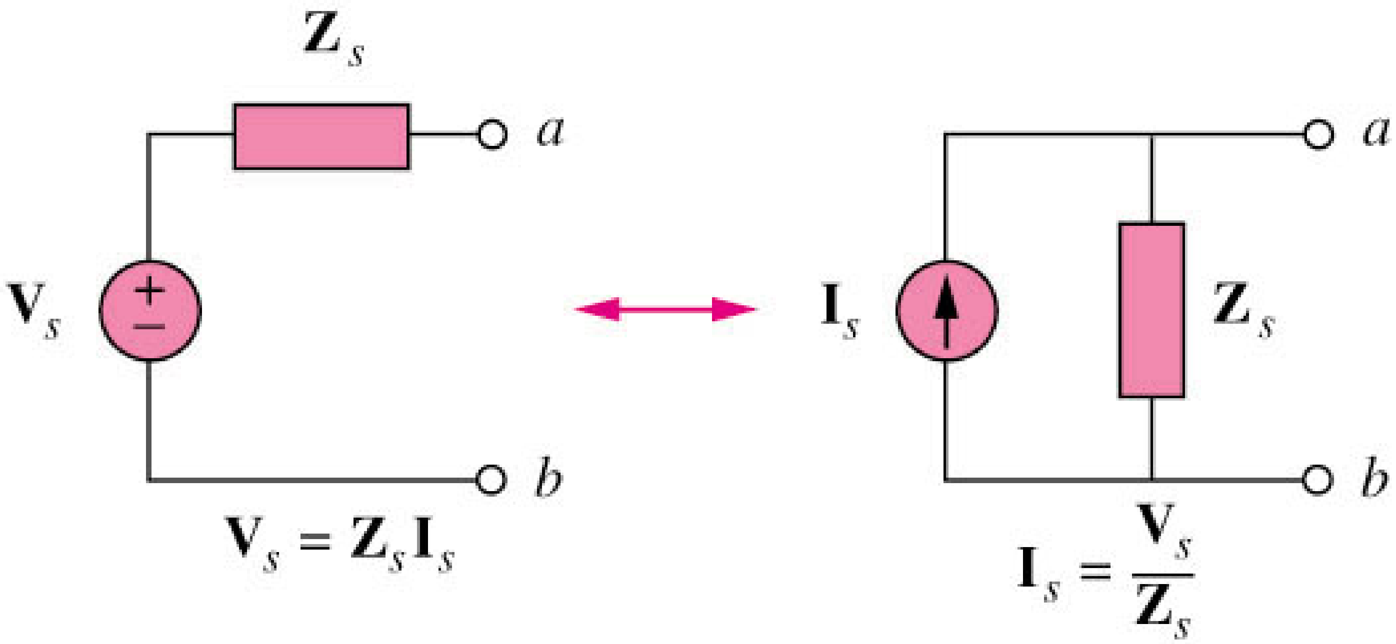
Example 3

Calculate v_o in the circuit of figure shown below using the superposition theorem.



V_o = **4.631 sin(5t – 81.12°) + 1.051 cos(10t – 86.24°) V**

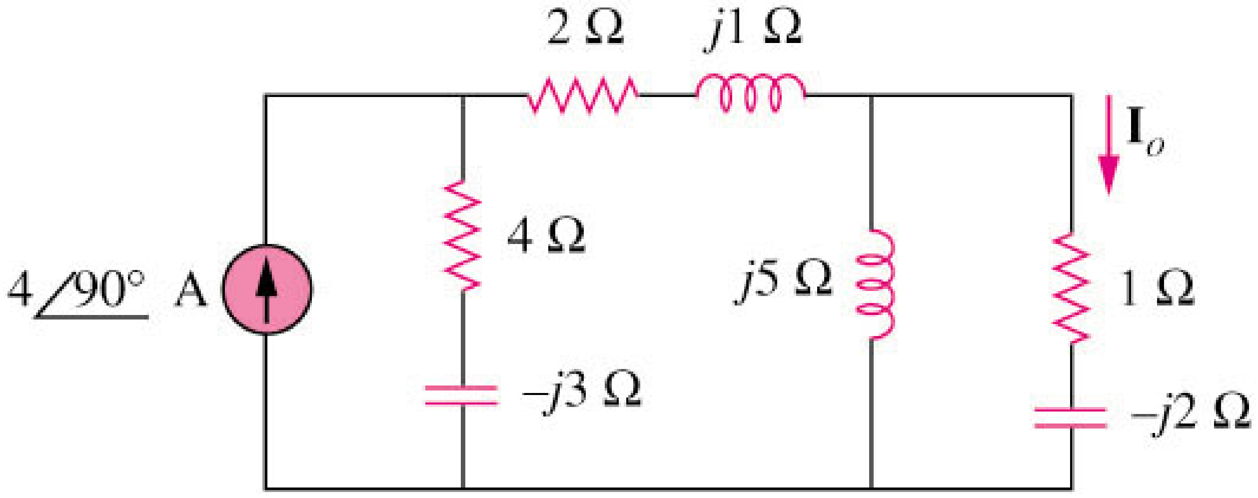
10.5 Source Transformation (1)



10.5 Source Transformation (2)

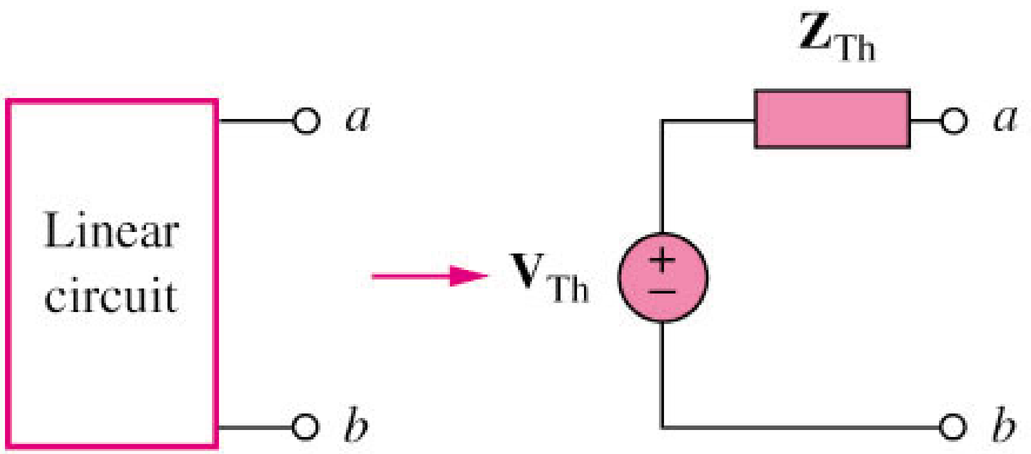
Example 4

Find Io in the circuit of figure below using the concept of source transformation.

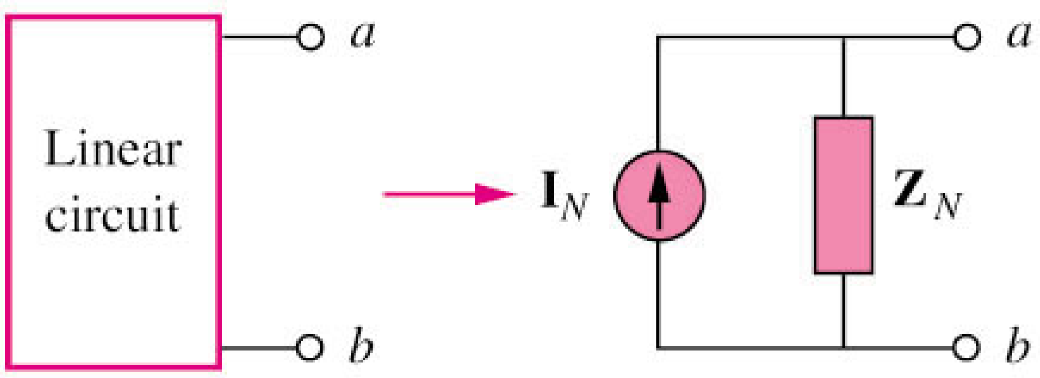


I_o = **$3.288\angle 99.46^\circ$ A**

10.6 Thevenin and Norton Equivalent Circuits (1)



Thevenin transform

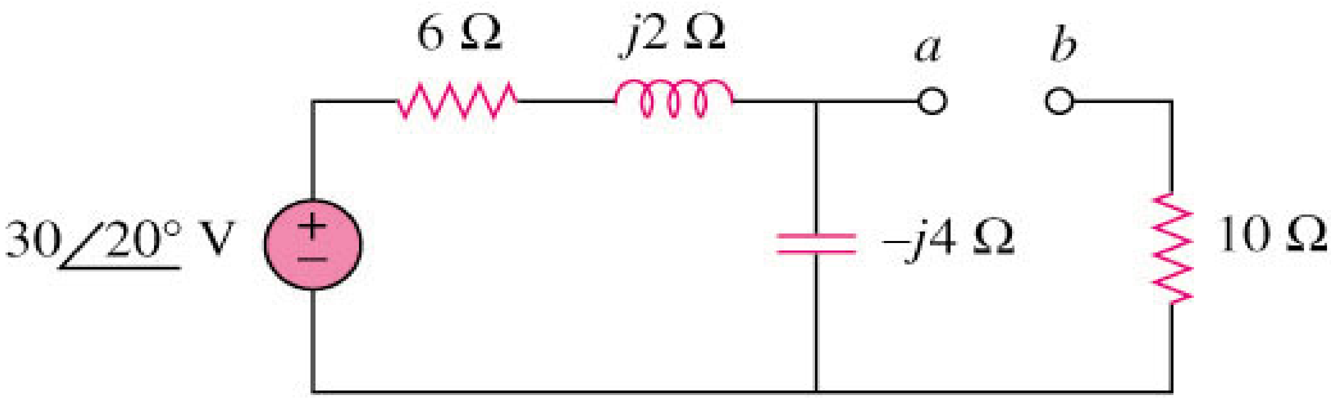


Norton transform

10.6 Thevenin and Norton Equivalent Circuits (2)

Example 5

Find the Thevenin equivalent at terminals a–b of the circuit below.



Z_{th} =**12.4 – j3.2 Ω**

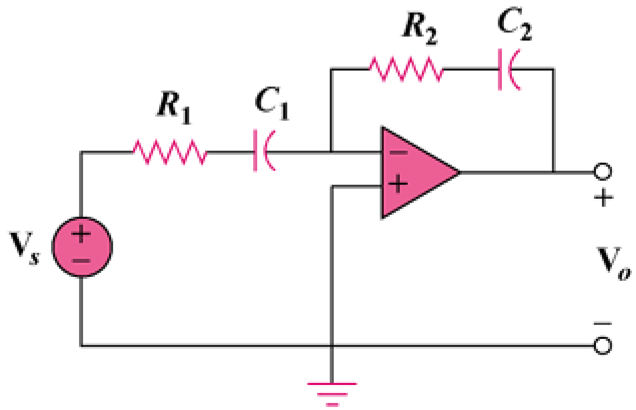
V_{TH} = **18.97∠-51.57° V**

HW16 Ch10: 40, 45, 49, 55, 72, 79

10.7 Op Amp AC Circuit

Example 5

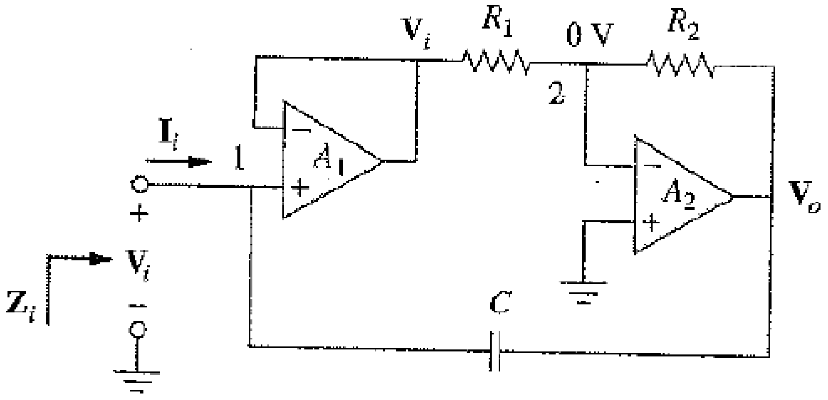
Evaluate the voltage gain $\mathbf{A}_v = \mathbf{V}_o/\backslash\mathbf{V}_s$ in the op amp circuit. Find $\mathbf{A}_v$ at $\omega = 0$, $\omega \to \infty$, $\omega = 1/R_1C_1$, and $\omega = 1/R_2C_2$.



10.9 Application

Capacitance Multiplier

Find the equivalent impedance Z_i



10.9 Application

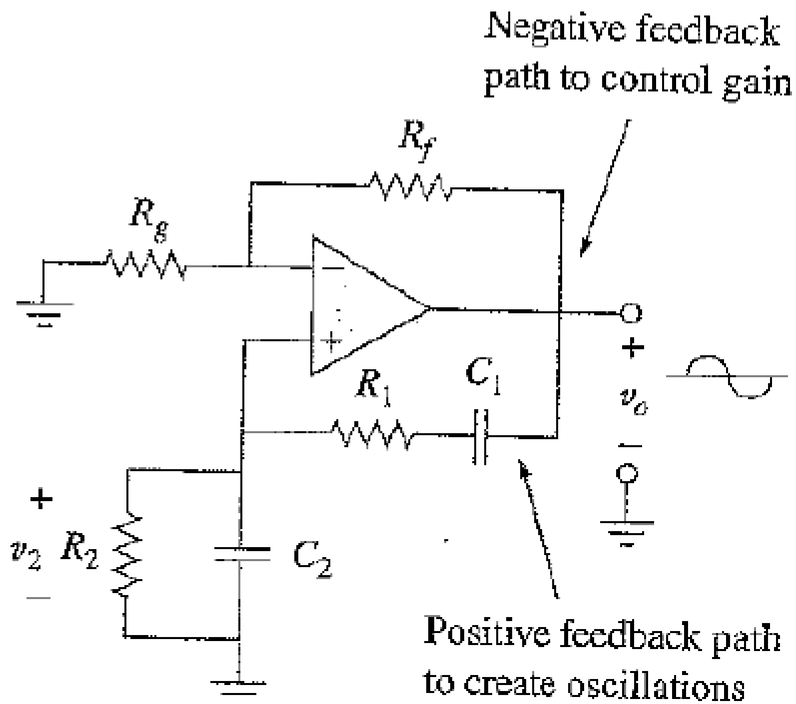
Oscillator

An oscillator is a circuit that produces an ac waveform as output when powered by a dc input.

Barkhausen Criteria

1- The overall gain of the oscillator must be unity or greater. Therefore, losses must be compensated for by an amplifying device.

2- the overall phase shift (from input to output and back to the input) must be zero.
V_o and V_2 should be in phase



Wien-bridge oscillator